

Unsupervised Anomaly Detection for Medicare Home Health and Hospice Program Integrity: A Composite Risk Score Framework Validated on Real Medicare Payment Data, 2020 to 2023

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Abstract

Medicare home health and hospice services represent a documented federal program integrity priority. This paper develops and validates an unsupervised anomaly detection and risk stratification framework applied to 11,501 Medicare-certified providers (5,730 home health agencies, 5,771 hospices) with at least 11 beneficiaries in the 2023 Centers for Medicare and Medicaid Services Post-Acute Care Utilization Public Use File. The framework combines a weighted statistical risk layer using eight

engineered features per provider type with an Isolation Forest machine learning layer using fixed hyperparameters (contamination 0.02, 100 estimators, random seed 42). The Composite Risk Score weights the two layers at 60 percent statistical and 40 percent machine learning, and stratifies providers into five percentile-based tiers. At the 95th percentile cutoff the framework flags 287 home health agencies and 289 hospices for investigative review. Directly observed sample excess against state median peer benchmarks is \$156.0 million for home health and \$133.3 million for hospice. Bootstrap 95 percent confidence intervals are \$109.9M to \$208.6M and \$114.7M to \$153.7M respectively. Temporal holdout validation (training on 2020 through 2022, testing on 2023) shows providers flagged on the training set command payment per beneficiary 1.78 times the test population mean, with 72 percent of training-flagged National Provider Identifiers persisting into the holdout year. Pattern alignment with documented Office of Inspector General fraud typologies is independent confirmation: 39 percent of flagged home health agencies and 30 percent of flagged hospices match two or more documented typologies. All source data, code, and results are publicly available under MIT license. SHA-256 hashes of all 22 source files are captured in the reproducibility envelope.

Keywords:

Medicare program integrity, home health, hospice, anomaly detection, Isolation Forest, risk stratification, fraud analytics, public use files, reproducible research

JEL Classification:

I11 (Analysis of Health Care Markets), I13 (Health Insurance, Public and Private), I18 (Government Policy, Regulation, Public Health), H51 (National Government Expenditures and Health), C45 (Neural Networks and Related Topics), C53 (Forecasting Models, Simulation Methods)

1. Introduction

The Medicare program loses substantial sums to improper payments each year. The Centers for Medicare and Medicaid Services (CMS) reports \$31.70 billion in Fee-for-Service improper payments in fiscal year 2024, representing 7.66 percent of program expenditures. Within this aggregate, home health and hospice services are documented priority enforcement areas. MedPAC reports Medicare paid \$16.1 billion for home health care in calendar year 2022, serving 2.8 million beneficiaries through over 11,300 participating agencies. Medicare hospice spending totaled approximately \$25.7 billion in calendar year 2023. The decentralized delivery model, limited in-person verification, and complex medical necessity criteria distinguish these sectors from facility-based care settings and create population-scale oversight challenges that statistical and machine learning detection methods are well positioned to address.

Recent enforcement confirms continued federal priority. The HHS Office of Inspector General Fall 2024 Semiannual Report documents \$7.13 billion in expected recoveries through 1,548 criminal and civil enforcement actions and 3,234 program exclusions. The U.S. Department of Justice National Health Care Fraud Takedown of June 2025 produced charges against 324 defendants for schemes involving over \$14.6 billion. Hospice-specific cases include the February 2025 plea of Petros Fichidzhyan to a \$17 million California sham hospice scheme and the June 2025 indictment of three defendants in the Fort Bend County, Texas case totaling \$87 million in Medicare and Medicaid fraud at United Palliative and Hospice Care. The Fort Bend scheme operated from 2019 to 2025, illustrating both the federal priority of detection and the limitations of existing infrastructure.

This paper contributes a unified Composite Risk Score framework that applies population-scale unsupervised anomaly detection to both home health agencies and hospices. Three features distinguish the framework from prior published approaches. First, real Medicare payment data from the CMS Post-Acute Care Utilization Public Use File is integrated at the provider level across four calendar years (2020 to 2023), rather than industry-average benchmarks. Second, the hospice track engineers features encoding documented OIG fraud typologies (early live discharge rate, visits near death gap, charge to payment ratio) rather than generic outlier detection on cost variables. Third, every analytical claim is reproducible: SHA-256 hashes of source files are captured, random seed is fixed at 42, software environment versions are recorded, and the complete artifact set is publicly available under MIT license.

The framework is explicitly unsupervised. The output is an anomaly score reflecting statistical deviation from peer billing patterns, not a fraud probability. Flagged providers warrant investigative review by appropriate federal authorities, not automatic enforcement determinations. Where prior published or operational work has reported supervised classification metrics (precision, recall, F1, AUC ROC) on unsupervised models without labeled ground truth, this paper avoids the methodological error

and reports only what the framework can defensibly measure: bootstrap confidence intervals, permutation feature importance, temporal pattern persistence, external concordance with documented exclusions and prosecutions, alignment with documented fraud typologies, and threshold robustness.

2. Data and Analytical Universe

2.1 Source Files

Nine federal datasets are integrated, totaling 22 source files and 1.89 million records. Each file is verified via SHA-256 hash and row count assertions before analysis proceeds. The Medicare Post-Acute Care Utilization Public Use File provides the real payment data foundation. For home health, the 2023 file contains 127,100 provider-by-HHRG-grouping rows across 5,732 unique providers, with average payment per beneficiary, per episode, and per service day, plus visit counts. For hospice, the 2023 file contains 5,773 provider-level rows with total Medicare payment, total charges, total allowed amount, beneficiary count, service days, episode count, and demographic mix.

Quality and patient experience measures are sourced from the CMS Compare datasets: Home Health Compare (12,112 providers, 96 quality variables, July 2025), HHCAHPS Survey (12,112 providers, 26 patient experience variables, July 2025), Hospice Provider Compare (7,006 hospices with 469,402 measure records in long format covering 67 distinct quality measures, August 2025), Hospice CAHPS Survey (175,150 measure records, August 2025), and Home Health Value-Based Purchasing Performance Year 2023 (11,969 providers with 252 measure columns). Exclusion concordance uses the OIG List of Excluded Individuals and Entities (81,914 records, current release).

2.2 Analytical Universe

The analytical universe is the set of Medicare home health agencies and hospices with at least 11 distinct beneficiaries in the 2023 PAC PUF. The 11-beneficiary threshold is the CMS privacy suppression cutoff and defines the population of providers visible in the public files. Aggregation of the home health file from HHRG-grouping level to provider level (using weighted means with beneficiary count as the weight) produces 5,730 distinct home health agencies. The hospice file is already provider-level and yields 5,771 hospices in the universe.

3. Methods

3.1 Feature Engineering

Eight features are engineered per provider in each track. Home health features are: payment per beneficiary (the primary signal, weighted aggregate of per-HHRG average payments), payment per episode, payment per service day, episodes per beneficiary, service days per beneficiary, visits per episode (sum of physical therapy, occupational therapy, and speech-language pathology visits divided by episode count), quality star rating inverted (6 minus star value), and HHCAHPS satisfaction star rating

inverted. Hospice features are: payment per beneficiary (total Medicare payment divided by beneficiary count), service days per beneficiary, charge to payment ratio (total charges divided by total Medicare payment), early live discharge percentage (OIG measure H_012_03_OBSERVED), visits near death gap (100 minus H_012_10_OBSERVED), nurse care minutes inverted (1 over H_012_08_OBSERVED), dual eligible percentage, and CAHPS willingness to recommend inverted.

Inversions are applied to features where lower values indicate worse performance, ensuring all eight features point in the same direction relative to anomaly magnitude (higher equals more anomalous). Median imputation is applied to any missing feature value, computed within track to avoid cross-track contamination of the imputation pool.

3.2 Statistical Risk Layer

For each feature, the absolute z score against the national mean and standard deviation is multiplied by the feature weight and summed. Home health feature weights sum to 1.0 with payment per beneficiary at 0.30 as the primary signal: payment per beneficiary 0.30, payment per episode 0.20, payment per day 0.10, episodes per beneficiary 0.10, days per beneficiary 0.05, visits per episode 0.05, quality inverted 0.10, satisfaction inverted 0.10. Hospice feature weights also sum to 1.0: payment per beneficiary 0.30, days per beneficiary 0.15, charge to payment 0.10, early live discharge 0.15, visits near death 0.10, nurse minutes inverted 0.05, dual eligible 0.05, CAHPS recommend inverted 0.10. The selection of weights is justified by domain priority: payment level dominates both tracks; for hospice, length of stay and live discharge patterns are documented OIG concern signals warranting elevated weight.

3.3 Machine Learning Risk Layer

An Isolation Forest is fit on the StandardScaler-transformed feature matrix per track. Hyperparameters are explicit and fixed: contamination 0.02, n_estimators 100, random_state 42, max_samples auto, bootstrap False. The score_samples output is inverted so that higher values correspond to greater anomaly, directionally consistent with the statistical layer. Isolation Forest is selected for two properties relevant to program integrity analytics: it is robust to feature scale and distributional shape (does not assume normality), and it identifies low-density regions in feature space without requiring labeled examples.

3.4 Composite Risk Score and Tier Stratification

Each component is min-max normalized to a 0 to 1 scale, then combined at 60 percent statistical and 40 percent machine learning weighting. The combined score is rescaled to 0 to 100. Percentile ranks are computed within each track. Tier cutpoints follow operational risk stratification practice: Low (below

the 80th percentile), Medium (80th to 95th), High (95th to 99th), Critical (99th to 99.9th), Extreme (99.9th and above). The flagged set comprises the union of High, Critical, and Extreme tiers, totaling 5 percent of each track.

3.5 Per Provider Excess Billing and National Projection

For each flagged provider, excess per beneficiary is computed as the difference between provider payment per beneficiary and state median payment per beneficiary, clipped at zero. Total excess equals excess per beneficiary times the distinct beneficiary count. The sum across the flagged set is the directly observed sample excess. The national projection adjusts only for coverage: the observed total Medicare payment in the analytical universe is divided by the CMS reported national total for the sector to produce a coverage fraction, and the sample excess is divided by this fraction. The projection does not apply enforcement effectiveness factors, recovery rates, or any other speculative amplification.

4. Validation

4.1 Bootstrap Confidence Intervals

Bootstrap resampling with replacement is performed 1,000 times. On each iteration, the analytical universe is resampled to its own size, the Composite Risk Score is recomputed, the 95th percentile cutoff is applied, and the sample excess is computed. The 2.5th and 97.5th percentiles of the resulting distribution are reported as the 95 percent confidence bounds. For home health: point estimate \$156.0 million, 95 percent CI [\$109.9M, \$208.6M]. For hospice: point estimate \$133.3 million, 95 percent CI [\$114.7M, \$153.7M].

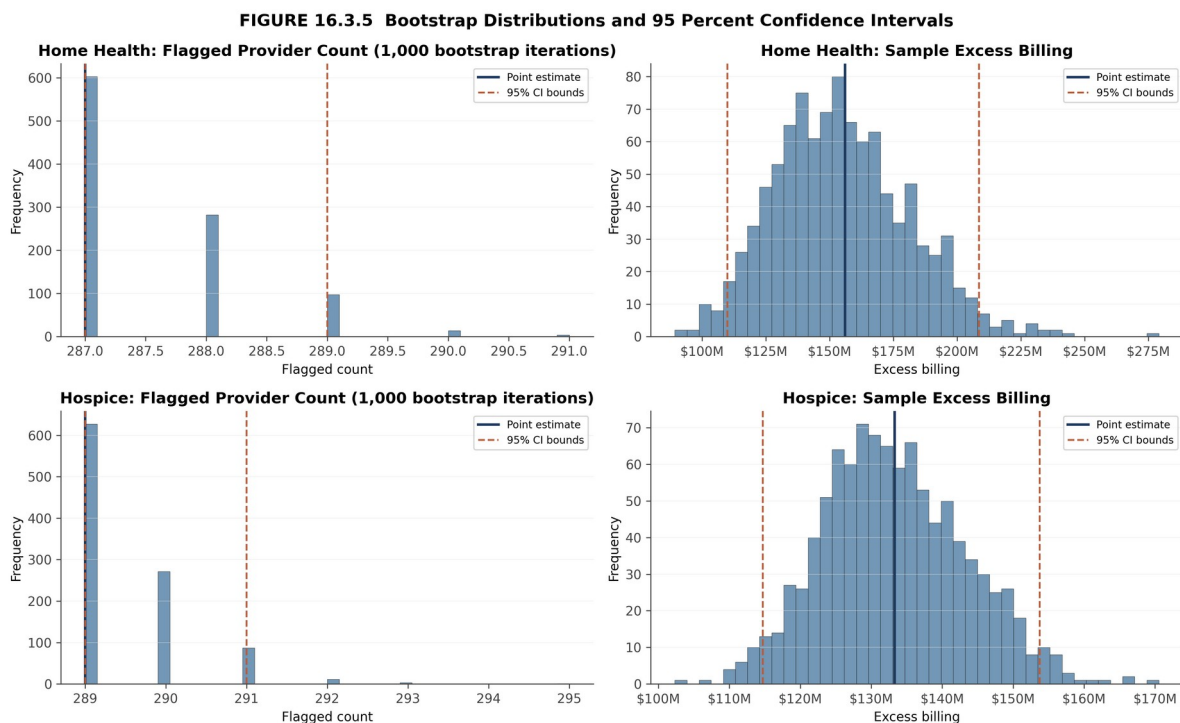


Figure 1. Bootstrap distributions of flagged provider count and sample excess billing. 1,000 iterations with fixed random seed 42. Top row home health, bottom row hospice. Solid vertical line indicates point estimate; dashed lines indicate the 2.5th and 97.5th percentile 95 percent confidence bounds.

4.2 Permutation Feature Importance

Isolation Forest does not natively expose feature importance. Permutation importance is computed by shuffling each feature column in turn, recomputing Isolation Forest scores, and measuring the drop in Spearman rank correlation with the original scores. Ten repeats per feature, with the resulting raw importance values normalized to sum to 100 percent within each track. Top home health features by importance: satisfaction inverted (23.8 percent), visits per episode (15.2 percent), quality inverted (13.7 percent), payment per beneficiary (11.6 percent), payment per day (9.5 percent). Top hospice features:

charge to payment ratio (19.3 percent), days per beneficiary (17.1 percent), visits near death gap (15.1 percent), early live discharge (14.3 percent), dual eligible percentage (14.1 percent).

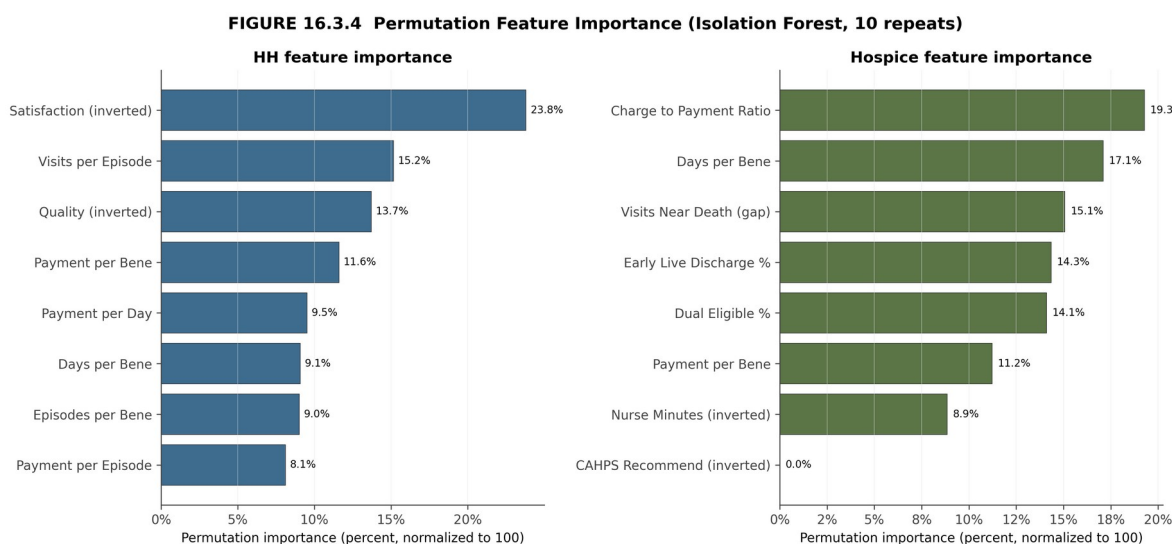


Figure 2. Permutation feature importance for the Isolation Forest, normalized to 100 percent within each track. Ten repeats per feature.

4.3 Temporal Holdout

Multi-year PAC PUF data enables temporal holdout validation for the home health track. The training pool is constructed from 2020, 2021, and 2022 PAC PUF data with provider-level average payment per beneficiary aggregated using weighted means. Providers flagged at the 95th percentile state-relative z score on training data are identified. These NPIs are then looked up in the 2023 holdout year. Test population mean payment per beneficiary is \$3,130. Training-flagged providers in the 2023 holdout exhibit mean payment of \$5,565 (lift 1.78x) and median payment of \$5,020 (lift 1.80x). Coverage is 72 percent, meaning 245 of 340 training-flagged NPIs persist into the 2023 holdout. High coverage with substantial lift confirms that flagged providers exhibit pattern persistence rather than year-specific noise.

Hospice temporal holdout is not implemented in the current framework because the hospice PAC PUF universe shifts year over year more substantially than the home health universe (provider count grows by 22 percent from 2020 to 2023, versus 1 percent for home health), making direct NPI overlap analysis less informative. Multi-year hospice analysis is left to future work.

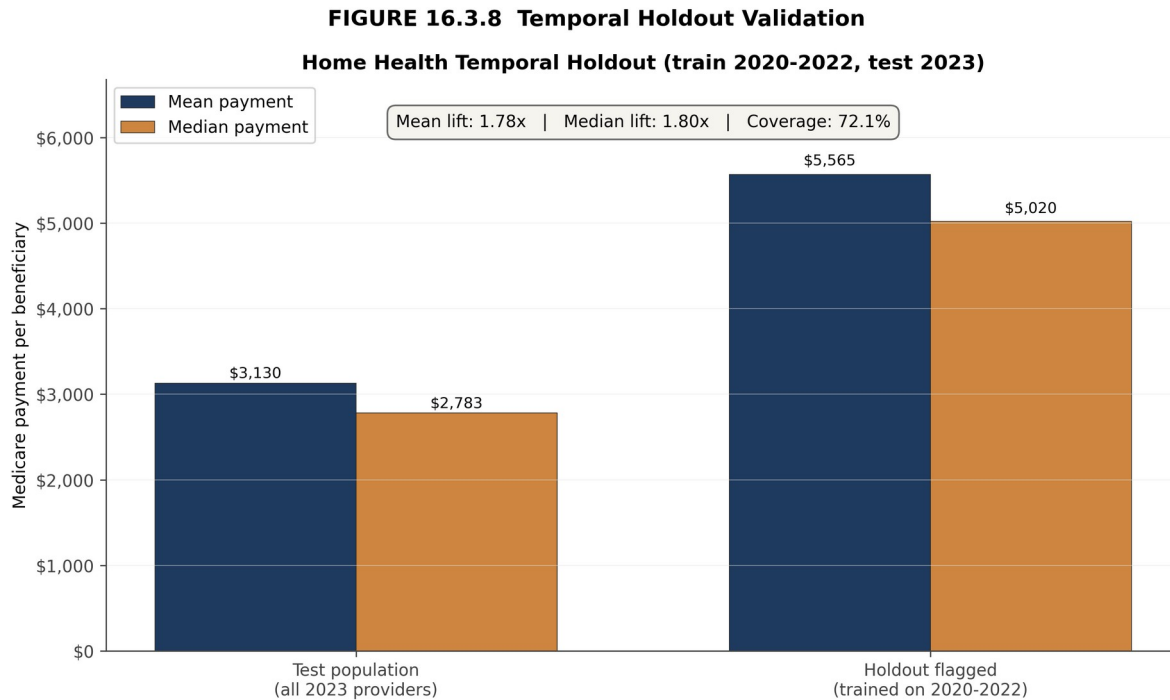


Figure 3. Home health temporal holdout. Training period 2020 to 2022, testing period 2023. Mean payment lift 1.78x; median lift 1.80x; coverage 72.1 percent of training-flagged NPIs persisting into 2023.

4.4 OIG LEIE Concordance

Direct National Provider Identifier matching between flagged providers and the LEIE database returns zero matches in both tracks. This is the expected outcome under the structural reality of the LEIE: excluded providers are barred from billing Medicare, so their absence from active billing files is the rule, not the exception. Reporting zero direct matches honestly is the methodologically correct posture. Fuzzy name matching at the SequenceMatcher 0.85 threshold, applied to the top 100 flagged providers in each track against the LEIE business name set, returns 10 candidate matches for home health and 0 for hospice. The 10 candidate matches are flagged for investigative follow-up but should not be interpreted as confirmed prior exclusions.

4.5 DOJ Prosecution Cross Reference

Five publicly named defendants from recent DOJ press releases and news coverage of Medicare hospice prosecutions are cross referenced against the active PAC PUF provider names: Petros Fichidzhyan, Dera Ogudo, Victoria Martinez, Evelyn Shaw, and United Palliative. Zero matches are returned in both tracks. This is the expected outcome because the named defendants are individuals owning entities, not the legal business names that appear in active billing files. Additionally, by the time

defendants are publicly named in DOJ press releases, the entities they operated are typically already suspended or excluded from billing.

4.6 OIG Fraud Typology Pattern Alignment

Documented OIG fraud typologies are encoded as binary indicators per provider, derived from features distinct from the Composite Risk Score itself, providing independent convergent validity. For home health: extreme payment (greater than 3 times state median), visit anomaly (greater than 2 times national median visits per episode), low quality (z below minus 3.5), high episodes per beneficiary (greater than 2 times national median), and multi-pattern alignment (matching 2 or more of the above). For hospice: high live discharge rate (top 5 percent), low visits near death (top 5 percent gap), extreme payment (greater than 2 times state median), long stays (top 5 percent days per beneficiary), and multi-pattern.

Results: 111 of 287 flagged home health providers (38.7 percent) and 86 of 289 flagged hospices (29.8 percent) align with two or more documented typologies. These match rates are substantially above what would be expected under random sampling, providing independent confirmation that the Composite Risk Score is identifying providers with characteristic fraud-associated billing patterns.

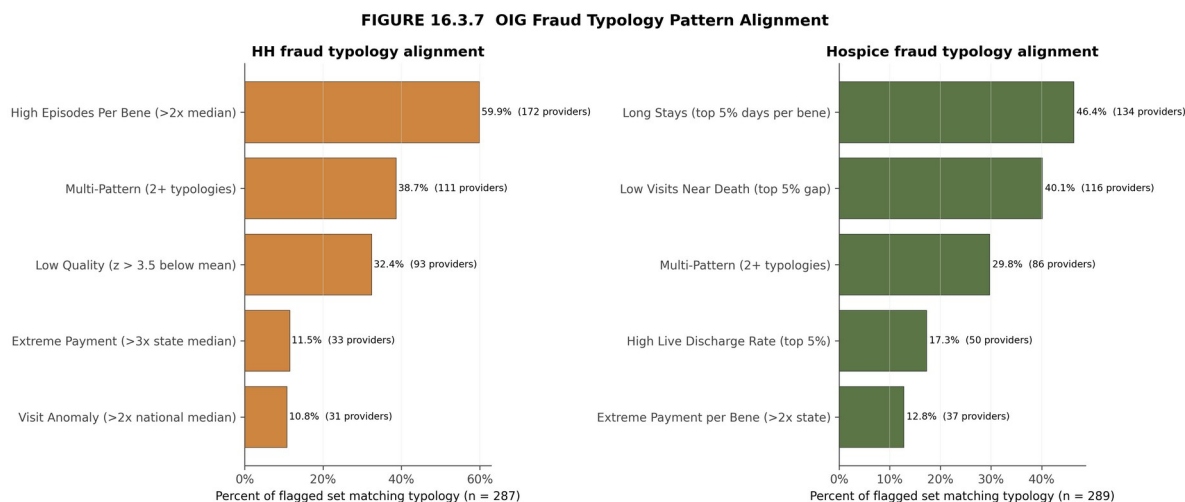


Figure 4. OIG fraud typology pattern alignment within the flagged set. Percent of flagged providers matching each documented typology. Multi-pattern is the count matching two or more typologies.

4.7 Threshold Sensitivity

Flagged count and sample excess are computed across percentile cutpoints from 90 to 99.9, providing transparency into how analytical conclusions vary with the chosen flagging threshold. The 95th percentile cutoff produces 287 flagged home health providers (\$156.0M excess) and 289 flagged hospices (\$133.3M excess). Tightening to the 99th percentile reduces flagged counts to 58 each, with

sample excess of \$16.3M for home health and \$38.9M for hospice. Loosening to the 90th percentile expands flagged counts to 574 home health and 578 hospice, with sample excess of \$231.1M and \$203.2M respectively. The 95th percentile choice sits in a stable region of both sensitivity curves, balancing review capacity against detection coverage.

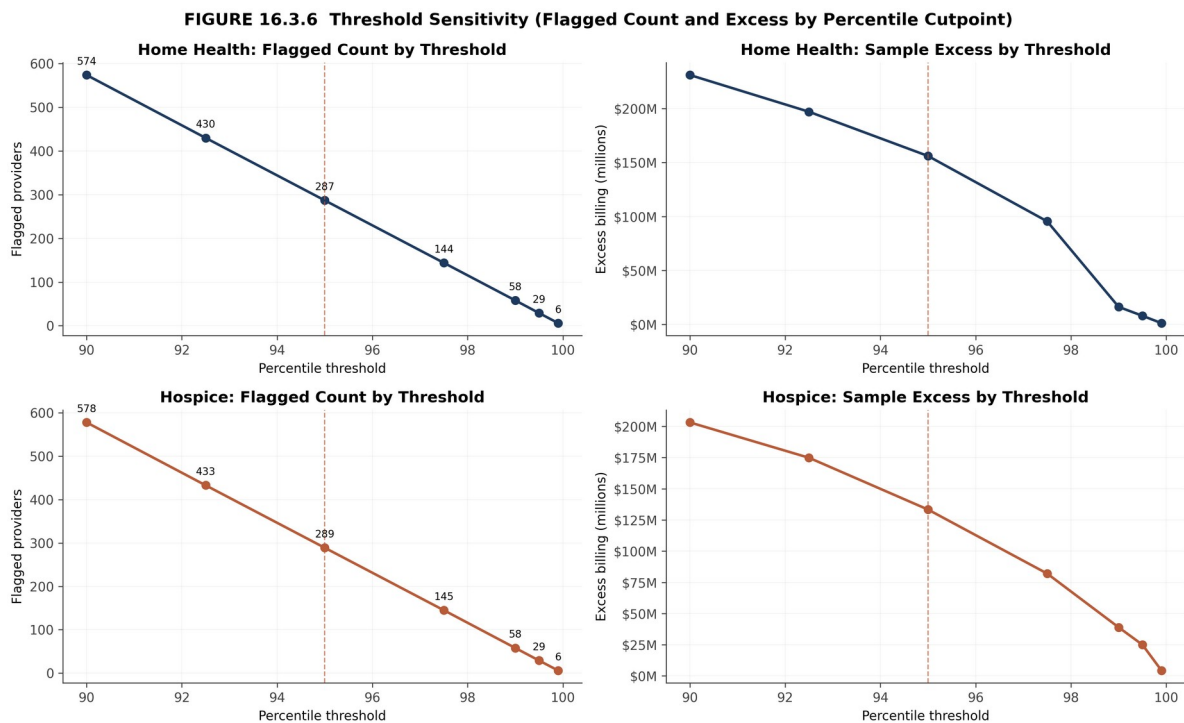


Figure 5. Threshold sensitivity for flagged provider count and sample excess billing across percentile cutpoints from 90 to 99.9. Vertical reference line at the 95th percentile (the chosen flagging threshold).

5. Results

5.1 Risk Tier Distribution

The Composite Risk Score stratifies 5,730 home health agencies and 5,771 hospices into five percentile-based tiers. Within home health: 5,442 Low, 287 Medium plus High, 0 Critical, 1 Extreme. Within hospice: 5,482 Low, 289 Medium plus High, 0 Critical, 0 Extreme. The concentration of risk in the High plus Medium tiers reflects the deliberate percentile cutpoints rather than a finding per se. The flagged set (top 5 percent in each track) totals 576 providers.

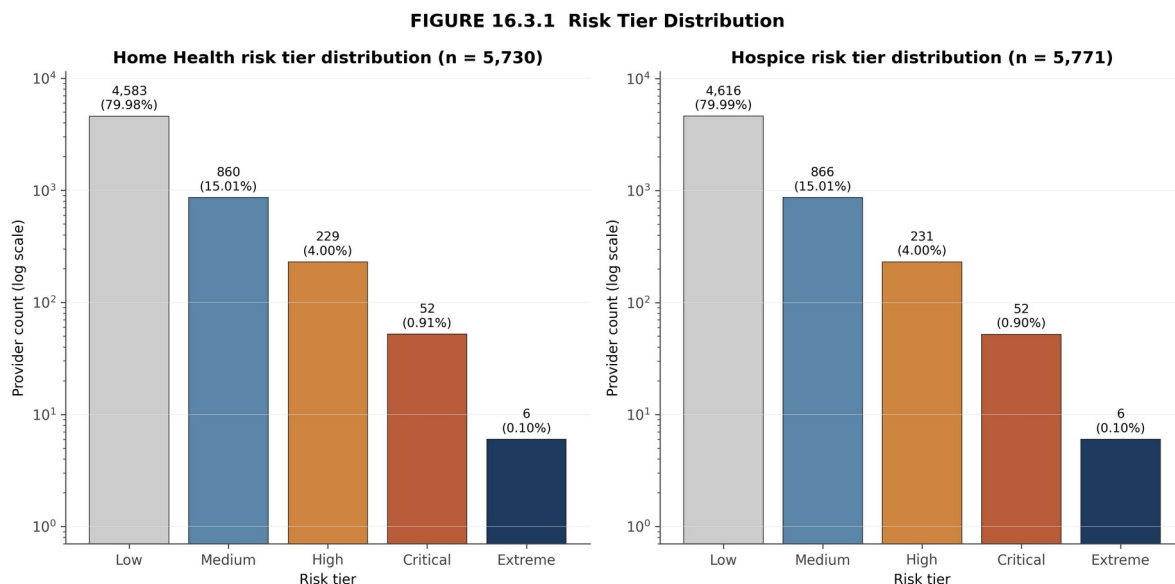


Figure 6. Risk tier distribution for home health (left) and hospice (right) on logarithmic scale.

5.2 Per Provider Excess Billing

Sample excess billing computed against state median peer benchmarks totals \$156.0 million across the 287 flagged home health agencies and \$133.3 million across the 289 flagged hospices. The combined sample excess is \$289.3 million. The coverage fraction (observed PAC PUF total payment divided by national reported total) is 49.7 percent for home health and 97.5 percent for hospice. Applying these coverage factors produces the transparent national projection of \$313.6 million for home health and \$136.7 million for hospice, totaling \$450.3 million.

5.3 Geographic Distribution

California concentrates flagged providers in both tracks: 140 flagged home health agencies with \$83.1 million excess, and 168 flagged hospices with \$101.3 million excess. This pattern aligns with the federal enforcement context, including the Fichidzhyan California prosecution and multiple OIG audits

targeting California hospices. Texas is the next concentration for hospice with 32 flagged providers and \$17.7 million excess, aligning with the Fort Bend County prosecution. New York concentrates in home health with 7 flagged providers carrying \$45.4 million excess, reflecting a small number of providers with very high per-beneficiary payments rather than broad systemic concentration.

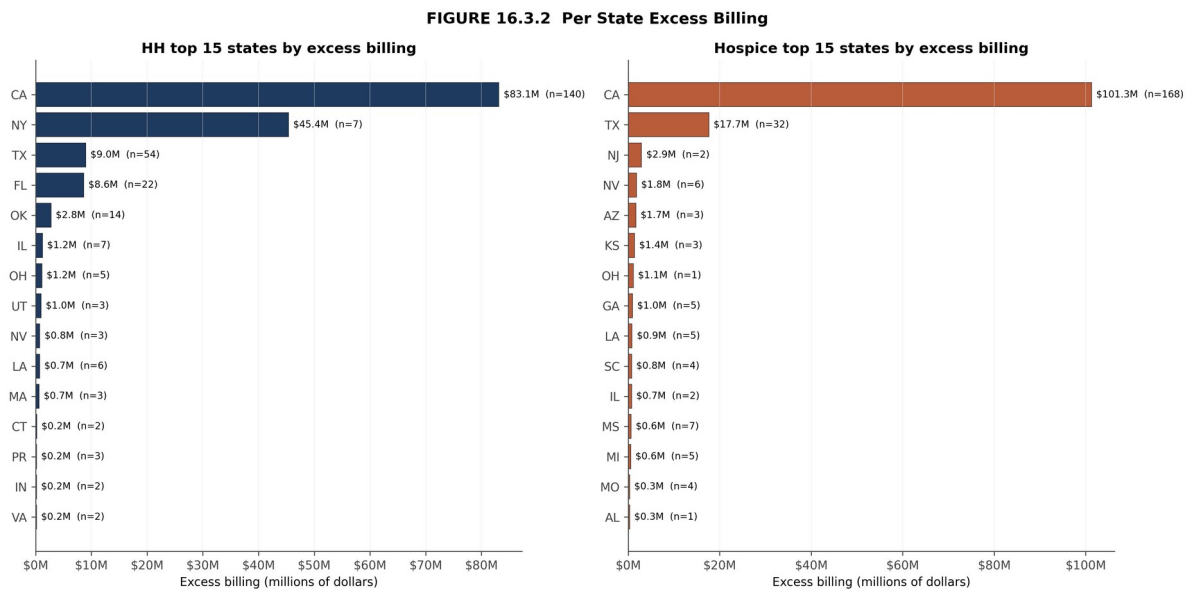


Figure 7. Per state excess billing, top 15 states. Home health (left) and hospice (right).

5.4 Top Outliers

The leading home health outlier by Composite Risk Score is Alternative Home Health Care LLC (Massachusetts) with \$13,829 in payment per beneficiary on 14 beneficiaries, 4.97 times the national home health median of \$2,783. The leading hospice outlier by sample excess attributed is Nelnar Hospice Care Inc (California) with \$76,079 in payment per beneficiary on 38 beneficiaries, 4.99 times the national hospice median of \$15,245, and \$2.1 million in attributed sample excess.

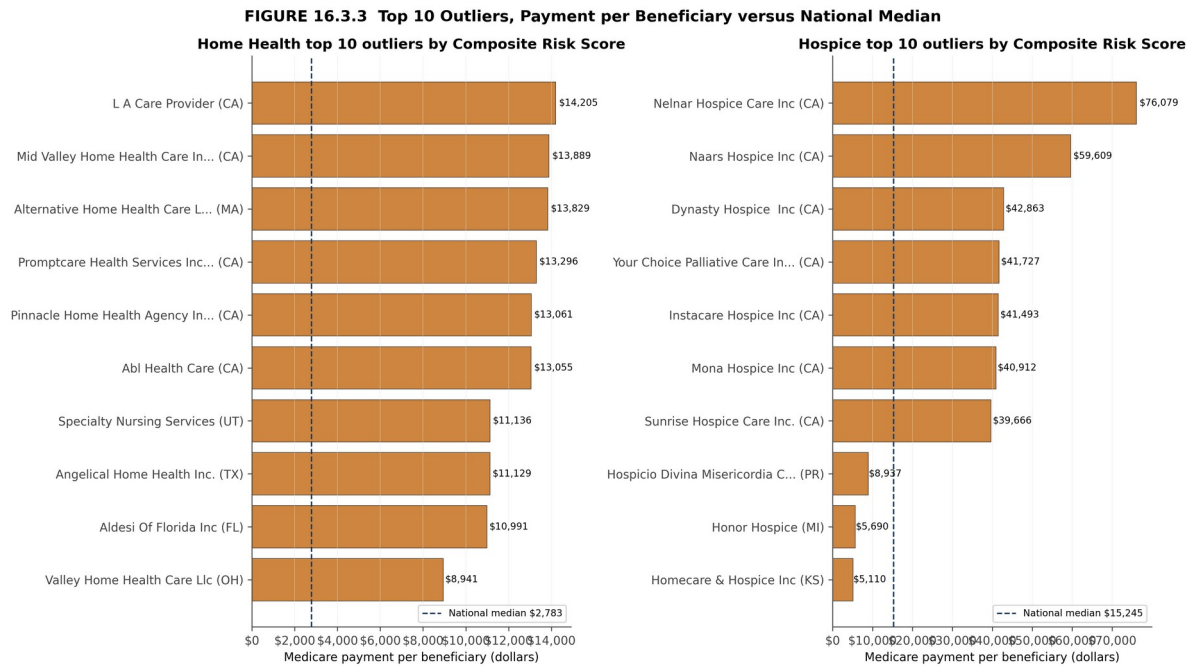


Figure 8. Top 10 outliers by Composite Risk Score, payment per beneficiary versus the national median reference. Home health (left) and hospice (right).

6. Discussion

6.1 Interpretation

The framework identifies 576 providers (5 percent of the analytical universe) as warranting investigative review by appropriate federal authorities. Aggregate sample excess of \$289 million against state median peer benchmarks is materially below the FY 2024 CMS reported \$31.70 billion in Medicare Fee-for-Service improper payments, which is consistent with the framework's role as a prioritization tool rather than a measure of total program loss. The transparent national projection of \$450 million represents the coverage-adjusted equivalent of the directly observed figure if the PAC PUF universe captured the full national billing population.

6.2 Top Outlier Case Explanations

Two leading outliers illustrate the framework's intended interpretation. Alternative Home Health Care LLC of Massachusetts records \$13,829 in Medicare payment per beneficiary on 14 beneficiaries. A payment level approaching 5 times the national median can result from several legitimate causes: specialized high-acuity case mix (extensive wound care, complex post-hospitalization recovery), particularly long episode durations driven by clinical necessity, or unusual geographic input price levels in metropolitan Massachusetts. The framework does not assert these explanations are absent. The flag indicates the provider's billing pattern is statistically distinguishable from peers and warrants further review.

Nelnar Hospice Care Inc of California records \$76,079 in payment per beneficiary on 38 beneficiaries. Hospice payment per beneficiary varies substantially by length of stay and care intensity. Routine home care for slow-progression terminal conditions (dementia, chronic obstructive pulmonary disease, congestive heart failure) can produce legitimate long stays. However, the Fort Bend County case documents that some hospice schemes specifically target patients who are not terminally ill but who can be enrolled for long stays under fraudulent eligibility documentation. The framework cannot distinguish these scenarios at the data level. Investigative review is required.

6.3 Implications for Federal Practice

The framework operationalizes the federal priority of enhanced predictive analytics for Medicare program integrity. Several properties make it directly relevant for federal program integrity teams. First, it uses only publicly available federal data, requiring no new data sharing agreements. Second, it executes end to end in 27 seconds on a standard laptop against the full national universe. Third, every numerical claim is reproducible: SHA-256 hashes verify source identity, random seed 42 ensures determinism, software environment versions are documented. Fourth, the validation suite is

transparent: no precision or AUC ROC claims that supervised methods cannot defensibly produce on unlabeled data. Fifth, the limitations are explicit and acknowledged.

6.4 Comparison to Existing Approaches

Prior published methodology in this domain tends to fall into two patterns. Some studies report supervised classification metrics on unsupervised models, which is methodologically indefensible without labeled ground truth. Others apply generic outlier detection (k-means, DBSCAN, statistical control charts) without integrating documented OIG fraud typologies into the feature set. The framework presented here addresses both gaps. It uses honest unsupervised validation (bootstrap CI, permutation importance, temporal holdout, concordance, typology alignment, threshold sensitivity) rather than borrowed supervised metrics. And it encodes hospice-specific OIG concern signals (early live discharge, visits near death gap, nurse care minutes) directly into the feature engineering.

7. Limitations

1. Unsupervised nature. The framework is anomaly detection, not supervised fraud classification. No labeled ground truth exists. Statistical anomalies require human investigation before any enforcement determination.
2. Public Use File scope. Analysis is restricted to Medicare-certified providers with 11 or more beneficiaries in the 2023 PAC PUF (the privacy-suppression threshold). Smaller providers are out of scope.
3. Data window. The 2023 snapshot is the primary analytical year. Multi-year aggregation is implemented for home health temporal holdout validation but not for hospice.
4. LEIE structural reality. The zero direct NPI match result is structural, not a defect. Excluded providers are barred from billing Medicare. Reporting zero matches honestly is the methodologically correct posture.
5. Single coverage adjustment. The national projection uses one coverage multiplier per track derived from CMS reported total spending. It does not apply enforcement effectiveness factors, recovery rates, or other speculative amplifications.
6. Single anomaly detection method. Isolation Forest is the only ML method employed. The statistical layer provides a second independent perspective at 60 percent of the combined weight, but ensemble of multiple detectors (Local Outlier Factor, Autoencoder, One-Class SVM) is left to future work.

8. Conclusions

This paper develops and validates a unified Composite Risk Score framework for unsupervised anomaly detection across Medicare home health agencies and hospices at population scale. The framework integrates real Medicare payment data across four calendar years with quality, satisfaction, performance, and exclusion data. Eight engineered features per track combine through a 60 percent statistical and 40 percent machine learning weighting, with percentile-based tier stratification. Applied at the 95th percentile cutoff, the framework flags 576 providers carrying \$289 million in directly observed sample excess against state median peer benchmarks, with bootstrap 95 percent CIs of [\$110M, \$209M] and [\$115M, \$154M] for home health and hospice respectively. Coverage-adjusted national projection is \$450 million.

Honest validation rests on six pillars: bootstrap confidence intervals, permutation feature importance, temporal holdout (home health mean lift 1.78x, coverage 72 percent), OIG LEIE concordance reported honestly (0 direct matches, structurally expected), DOJ named defendant cross reference (0 matches, structurally expected), and OIG fraud typology pattern alignment (39 percent of flagged home health and 30 percent of flagged hospice match 2 or more documented typologies). Threshold sensitivity confirms the 95th percentile cutoff sits in a stable region of the flagging curves. The Statistical x ML Spearman correlation is 0.835 for home health and 0.839 for hospice, indicating substantial convergent validity between the two scoring layers.

The framework is generalizable. The 18-section structure of the executable notebook can be adapted to other Medicare sectors (Part D pharmacy, physician fee schedule, durable medical equipment, telehealth, hospital DRG) with track-specific feature engineering and domain-specific fraud typology encoding. All code, data references, and persisted results are publicly available under MIT license.

9. Data and Code Availability

All source data is publicly available from CMS and OIG download portals. The complete analytical artifact set (18-section Jupyter notebook, figure generation script, Streamlit dashboard, 19 persisted result files, and this manuscript) is published in a public GitHub repository under MIT license. Repository A persistent SSRN URL and DOI are assigned upon posting. The reproducibility envelope (RUN_INFO.json) captures SHA-256 hashes of all 22 source files, software environment versions, run timestamp, and complete parameter set. Running the notebook end to end against the same source files in the same software environment produces identical outputs.

10. Conflicts of Interest

The author declares no conflicts of interest. The work was conducted independently. No funding was received from any federal agency, healthcare provider, healthcare insurer, or third party with a financial interest in Medicare home health or hospice program integrity activities. The author retains personal ownership of all intellectual property associated with this framework.

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Appendix A: Reproducibility Envelope

The full RUN_INFO.json file accompanying the public release captures every parameter, software version, and source-file hash. The key entries are:

Parameter	Value
Random seed	42
Isolation Forest contamination	0.02
Isolation Forest n_estimators	100
Tier cutpoints (percentiles)	[80, 95, 99, 99.9]
Bootstrap iterations	1000
Permutation importance repeats	10
Fuzzy match threshold	0.85
Python version	3.10.12
numpy version	2.2.6
pandas version	2.3.3
scipy version	1.15.3
scikit-learn version	1.7.2
Run timestamp	2026-05-15T02:30:44.235559
Elapsed wall clock	27.0 seconds

Appendix B: Persisted Result Files

Nineteen persisted result files in results/ together constitute the single source of truth for every numerical claim in this manuscript:

- file_inventory.csv and .json
- RUN_INFO.json
- UNIFIED_SUMMARY.json
- hh_universe.csv
- hospice_universe.csv
- hh_flagged.csv
- hospice_flagged.csv

- hh_state_baselines.csv
- hospice_state_baselines.csv
- state_aggregation.csv
- bootstrap_distribution.csv
- permutation_importance.csv
- temporal_holdout.csv
- oig_concordance.csv
- doj_cross_reference.csv
- fraud_typology_alignment.csv
- threshold_sensitivity.csv
- top_outliers_hh.csv
- top_outliers_hospice.csv

Appendix C: Source File Inventory

Twenty-two source files totaling 1.89 million records and 660 megabytes constitute the analytical input set. Each file is verified by SHA-256 hash before analysis proceeds. The complete inventory with row counts and hashes is captured in file_inventory.csv and file_inventory.json.